Sahaj Bhadja

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Summary

Full-stack Software Engineer with 2+ years shipping AI products end to end, building **LLM** pipelines that automate evaluation at scale, **retrieval** over 50K+ documents, and scalable **Python** and **TypeScript** platforms with high ownership.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Architected a multi-tenant AI platform on Azure SQL and Blob Storage with row-level security and normalized schemas for 500+ users, cutting dashboard latency from 1.2s to 350ms through query tuning.
- Rebuilt async **LLM orchestration** with FastAPI workers, explicit job states, and failure isolation, tripling throughput across 1K+ AI jobs under concurrent load while integration pipelines held 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built an **AI evaluation** product that auto-generates rubrics and scores text and handwritten submissions with GPT-4o, Claude, and Qwen-VL at 90%+ consistency, replacing manual grading for 150+ instructors.
- Scaled the platform to 10K+ daily requests with concurrent batch processing of 1K+ files, provider fallback routing, and per-call **cost tracking**, cutting feedback turnaround 60% on React and Flask.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Automated cross-platform data-sharing workflows on a **microservice architecture** that cut manual data entry 35%, owning work from requirement analysis through deployment and support with Microsoft and GitHub stakeholders in Agile sprints.
- Optimized large-scale API integrations with caching and pagination, improving response times 30%, and led an AWS Glue and S3 **ETL** redesign that boosted pipeline efficiency 68% and cut storage costs 15%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a TypeScript, React, and Node.js **CRM** with lead tracking, customer profiles, interaction history, and email and SMS notifications, synchronizing 2M+ records across locations in real time via WebSockets.
- Used Google Analytics on funnel drop-off points and feature usage to drive product changes that raised **lead conversions** 15% and cut resource wastage 42%, shipping via Jenkins CI/CD.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst – Research Fellowship *India*

- Engineered 21 behavioral **features** such as IP presence, URL length, and suspicious keywords over a 650K+ URL dataset, building an end-to-end classification pipeline for phishing and malware detection.
- Trained and compared Gaussian Naive Bayes, Random Forest, LSTM, Bi-LSTM, GRU, and CNN classifiers, reaching 96%+ detection **accuracy** and reporting actionable insights into emerging phishing and malware trends.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, C++
AI & ML: LLM Orchestration, Prompt Chaining, RAG, Embeddings, FAISS, Pinecone, Claude, GPT-4o, PyTorch, Feature Engineering, Classification
Full Stack: React, Next.js, Node.js, FastAPI, Flask, REST APIs, WebSockets, Microservices
Data & Infra: PostgreSQL, MongoDB, Supabase, Azure SQL, ETL Pipelines, AWS, Azure, Docker, CI/CD
Practices: System Design, Claude Code, Rapid Prototyping, Code Reviews, Git, Stakeholder Communication

Projects

AI Candidate Sourcing and Screening Agent | *Python, FastAPI, Claude, pgvector, Next.js, TypeScript*

- Designed an agent that embeds role requirements and public candidate profiles into pgvector, ranks matches with **hybrid retrieval**, scores fit against a rubric with Claude, and drafts **personalized outreach** in Next.js.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, Clerk, Convex*

- Built a semantic **retrieval** platform with transformer embeddings over FAISS and Pinecone across 50K+ documents, powering LLaMA-2 Q&A at 95% query accuracy and cutting document review time 60%.

Trend Tracker with Fashion Recommendation System Live | *CNN, React.js, Strapi, AWS EC2*

- Built and deployed a CNN-based **recommendation** model for an e-commerce trend tracker on React, Strapi, and AWS EC2, improving recommendation accuracy from 78% to 92% and lifting engagement across 3M+ users.